SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED
CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25
Question Count: 5

Code of Conduct
I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1) Cloud auto-Scaling and load balancing help companies handle unpredictable traffic spikes efficiently during seasonal sales.

Auto-Scaling:

It automatically increases or decreases Cloud resources based on demand. During high traffic, unnecessary servers are removed.

Load Balancing:

It distributes incoming user request across multiple servers to prevent any single server to prevent any single server from becoming overloaded.

Benefits:

- Reduces server overload and downtime.
- Improves application speed and Availability.
- Provides a smooth user Experience.
- Reduces costs by using resources only when needed.

Role of monitoring tools:

Monitoring tools track CPU usage, memory, response time, and incoming traffic.

Conclusion:

Auto-Scaling and load balancing improve Scalability, Availability and Customer Satisfaction while reducing operational costs.

2) Cloud storage allows Organizations to store large volumes of backup data securely, reliably, and Economically.

Durability and Data redundancy:

Durability: Ensures that stored data remains safe and available over time.

Data redundancy:

* Maintains multiple copies of data across different devices or locations
* If one storage device fails, data can be recovered from another copy.

Security measures:

* Encryption
* Access control
* Multi-factor Authentication.

Cost Advantages:

* Reduces the need to purchase physical devices.
* Eliminates much of the cost of maintaining an on-site data centre.

* Allows storage capacity to increase as required.

Disaster recovery Example:

If a company's local server is damaged by a fire, its encrypted cloud backups can be used to restore important files and databases.

Conclusion:

Cloud backup provides reliable data protection, security scalability and cost savings compared with traditional storage system.

3) A Content Delivery Network (CDN) is a group of geographically distributed servers that deliver website content & application data to users from nearby locations.

* Reduced Latency: Delivers data from servers closer to users.

* Edge servers: Store and deliver content near users, reducing loading time.

* Improved performance:

Reduces the workload on the main server and minimizes buffering.

* Global availability:

Cloud providers use multiple regions and data centres to improve service availability.

Ex: Netflix uses CDN Technology to deliver Video Content efficiently to users in different Countries, improving streaming quality and reducing buffering.

Conclusion:

CDNs improve Speed, reduce latency and provide a better user experience Worldwide.

4) Financial institutions must protect sensitive customer data and follow applicable regulatory requirements while using Cloud services.

Regulatory Compliance:

Cloud providers offers Security Controls and Compliance Support.

Logging:

Records user Activities and financial transactions for Auditing.

Monitoring:

Detects Suspicious Activities and possible Security threats.

Encryption:

Protects data during transmission and Storage.

Access Control:

Restricts Sensitive information to Authorized users.

Data Sovereignty:

Ensures data is stored in geographical regions permitted by applicable regulations.

Ex:

A bank uses Encryption, monitoring, and Audit logs to protect online fund transfers and identify suspicious transactions.

Conclusion:

Cloud security measures protect financial information and help institutions meet regulatory requirements.

5) Containerization Packages an Application and its dependencies into lightweight units called Containers.

Containers	Virtual Machines
Share the host OS Kernel	Run a separate guest OS.
Use fewer resources	Generally use more resources.
Start quickly	Usually start more slowly.
Support application portability.	Provide Complete Virtualized Environment.

Role of Kubernetes:

* Automates container deployment and management
* Scales applications accordings to demand.
* Restarts failed Containers.
* Distributes Workloads Across Servers.

Portability:

Containers help applications run consistently across different cloud environments, although some configuration changes may be needed.

Ex:

An online shopping application uses separate containers for login, payments, and product search. Kubernetes manages these services and scales them during high traffic.

Conclusion:

Containerization improves resource efficiency and portability, while Kubernetes simplifies deployment and scaling.